What can waves tell us?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a signal's travel from the motion of its source.

SUCCESS CRITERIA - I can...

1. Identify what a wave can carry.
2. Separate signal travel time from source speed.
3. Name evidence needed to interpret an observation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "wave" and "signal" to compare your model or result. Sentence starters are optional.

wave: a disturbance that can carry energy and information

signal: a disturbance that carries information

evidence: observations or measurements that support an idea

01 Bring back an earlier idea

Two equal-density spheres have radii r and 3 r. Supplied: area scales by 9, mass by 27.
Compare their area per mass.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: distance = speed × time
t = d/c for light in vacuum.
A signal takes time to travel. Receiving its light does not mean the source traveled with it.

Worked reasoning: A pulse travels 4 m along a rope in 2 s, so its average propagation speed is 2 m/s. A marked rope segment moves locally; it does not ride 4 m along with the pulse.

Your turn: Sketch three pulse positions and mark the same rope segment in each snapshot.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

A pulse travels along a rope while each marked rope segment moves mainly up and down. Does the material travel all the way with the pulse?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

04 Build the solution

A signal travels at a supplied speed 10 m/s across 20 m.
Find travel time using t=distance÷speed.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

What can waves tell us?

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

A student says seeing an asteroid’s light tells us exactly where it is “right now.” Explain one limitation.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A sample warms by 4°C after 600 J is transferred into it. **Explain** why the temperature rise and heat transfer need different units.

Word help: temperature = a measure of thermal state; it is not an energy amount

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

07 Exit check

Name one quantity a wave observation could help estimate and one additional piece of information needed.

Word help: estimate = find an approximate value

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Design a demonstration that distinguishes energy propagation from transport of the entire material. **Name** what you would track.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3